Time Series Analysis of COVID-19 Data using Python Libraries

Mayuukh Upadhyay,

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Calcutta University

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1. Abstract

This project presents a comprehensive time-series analysis of the global COVID-19 pandemic using the WHO daily reporting dataset, complemented by an application of unsupervised machine learning on structural pixel feature arrays. The primary goal was to extract actionable insights from raw temporal data by utilizing Python and the Pandas library for data wrangling, cleaning, and aggregation. Key findings include identifying the top five most affected nations, pinpointing the date of the highest global daily spike (January 30, 2022, with ~8.4 million cases), and visualizing the shifting regional epicenters through quarterly cross-tabulations. In addition to tabular data analysis, the project explored pattern recognition by applying the K-Means clustering algorithm to the MNIST digit dataset. The model successfully segmented the pre-flattened pixel-intensity vectors without prior label exposure, achieving a macro-averaged F1 score of 0.8628 after algorithmic label mapping. Ultimately, this internship project demonstrates a complete, end-to-end data science pipeline, bridging the gap between raw data ingestion and robust unsupervised performance evaluation.

2. Introduction

The exponential generation of data in the modern digital era has made data science an indispensable tool for understanding complex, large-scale events. This project focuses on analyzing the temporal and geographical progression of the COVID-19 pandemic, an event of unprecedented global relevance that requires rigorous statistical tracking. By leveraging Python—specifically industry-standard libraries such as Pandas, Scikit-Learn, and SciPy—this project involves the practical application of data manipulation, time-series analysis, and unsupervised clustering techniques.

The background material survey included studying the structure of the WHO global daily COVID-19 reports and the mathematical foundations of unsupervised clustering algorithms like K-Means. The procedure executed involved systematic data ingestion, temporal filtering (isolating the core pandemic timeframe from March 2020 to August 2023), executing high-level statistical aggregations, and culminating in the training and evaluation of a clustering model. The core purpose of this project was to solidify theoretical data science concepts through hands-on practice, successfully transitioning from handling structured tabular data to extracting structural variations from pre-flattened, high-dimensional numerical feature arrays.

Topics covered during the initial two-week training period:

- Python Programming Fundamentals
- Data Structures and their usage in Python
- Object Oriented Programming using Python
- Numpy and Matplotlib Libraries and its applications
- Pandas Library and its applications
- Scipy and Scikit Libraries and its application

- Fundamental Machine Learning Concepts
- Linear Regression and its ML Workflow
- Logistic Regression and its ML Workflow
- Single-layer and Multi-layer Perceptron and implementation
- DBScan as Density based ML algorithm and implementation
- K-Means Clustering and its implementation
- Model Evaluation Metrics (e.g., F1 Score, label mapping techniques)
- Large Language Models and their applications

3. Project Objective

- **Implementing a comprehensive data science workflow:** Executed the end-to-end pipeline encompassing data ingestion, cleaning, transformation, and analysis using Python.
- **Analyzing temporal pandemic trends:** Performed granular time-series analysis on the WHO dataset to identify peak infection dates, quarterly trends, and macro-level regional shifts.
- **Optimizing data processing efficiency:** Demonstrated proficiency in memory optimization and dataset management through targeted subsetting, boolean masking, and correct data type assignments.
- **Applying unsupervised machine learning:** Utilized K-Means Clustering on the MNIST dataset to independently partition pre-flattened numerical arrays based purely on intrinsic pixel-intensity coordinate distances.
- **Executing rigorous model evaluation:** Evaluated the performance of the unsupervised model against true labels by developing an algorithmic label-mapping loop and calculating the macro-averaged F1 score.

4. Methodology

4.1 Data Collection and Ingestion

The empirical foundation of this project relies on two distinct data structures:

- **Epidemiological Dataset:** The official WHO Global Daily COVID-19 dataset (WHO-COVID-19-global-daily-data.csv) was programmatically downloaded using the gdown library. This tabular dataset records daily country-level tracking metrics.

- **Pixel-Feature Dataset:** The MNIST handwritten digits dataset was imported directly via Scikit-Learn's load_digits module, yielding 1,797 samples. Rather than raw image files, this module provides pre-flattened, 64-dimensional numerical feature vectors representing the intensity values of an 8×8 grayscale grid.

4.2 Data Cleaning and Pre-processing Pipeline

To ensure analytical accuracy and computational efficiency, the raw data structures underwent a targeted processing pipeline:

- **Feature Isolation (Subsetting):** The raw COVID-19 dataset was filtered to extract an independent memory slice containing only five essential vectors: Date_reported, Country, WHO_region, New_cases, and New_deaths. This structural reduction minimized memory footprint using the .copy() method to avoid pointer issues.
- **Temporal Type Casting:** Raw date vectors, initially ingested as string objects, were cast into explicit Pandas datetime64 objects via pd.to_datetime(). This conversion enabled vectorized datetime accessor extraction (.dt).
- **Boolean Mask Temporal Slicing:** A chronological window bounding the core pandemic phases (**March 1, 2020, to August 31, 2023**) was established using bitwise AND (&) logical operations to extract a precise temporal subset.
- **Feature Processing Matrix:** Because the MNIST data is pre-packaged by Scikit-Learn as a clean matrix of numerical intensities, it required zero manual computer vision processing, image resizing, or noise filtering. The 64 feature columns were passed directly into the modeling pipeline.

4.3 Analytical and Machine Learning Modeling

- **Time-Series Resampling & Reshaping:** Temporal groupings were achieved using the .dt.to_period('Q') function to transform continuous daily logs into categorical quarterly bins. Spatial-temporal matrix indexing was finalized using pd.pivot_table(), setting WHO_region as index rows and monthly periods as feature columns.
- **Unsupervised Clustering Configuration:** The machine learning architecture utilized the K-Means algorithm from Scikit-Learn. The algorithm was configured with a fixed cluster hyperparameter ($K = 10$) to align with the known digit space (0–9). Centroid initializations were controlled with a fixed random_state=42 to ensure deterministic execution.
- **Algorithmic Cluster Validation & Mapping:** Because K-Means operates blindly without class labels, the output IDs are arbitrary. To bridge the unsupervised partitions with supervised target values (y), an automated mapping loop was built. For each isolated cluster array, the mathematical mode was computed via scipy.stats.mode(). This assigned each cluster index to its dominant ground-truth digit label before computing final validation statistics.

4.4 Repository Architecture

The full functional code base, source notebooks, and environment configurations have been compiled and pushed to a public version-control repository. The structured codebase can be reviewed at the following repository link: https://github.com/MayuukhU15

5. Data Analysis and Results

For machine learning, models a comparative analysis of the various models needs to be reported with the measured prediction accuracy levels.)

5.1 Descriptive Time-Series Analysis (COVID-19)

The final cumulative aggregations isolated the macro-level impact metrics across global boundaries. The analysis successfully identified the top five nations by cumulative infection counts:

Rank	**Country**	**Final Cumulative Case Metric**
1	United States of America	Highest Global Aggregate
2	China	Second Highest Aggregate
3	India	Third Highest Aggregate
4	France	Fourth Highest Aggregate
5	Germany	Fifth Highest Aggregate

Vectorized aggregation over the global tracking array determined that the absolute peak of the pandemic's transmission velocity occurred on **January 30, 2022**. On this single day, global daily new infections spiked to a historic maximum of **8,401,963 cases**, primarily driven by the worldwide emergence of highly transmissible variants.

5.2 Regional Shifting Trends (Pivot Aggregations)

By executing a multi-dimensional pivot table mapping, the geographical spread was analyzed on a monthly basis across specialized WHO regions (AMRO, EURO, SEARO, WPRO, EMRO, AFRO). The cross-tabulated matrix highlighted clear shifts in pandemic waves:

- Early-stage case concentrations dominantly occupied the European (EURO) and American (AMRO) sectors.
- Later temporal periods (specifically early 2022) exposed major case volume surges in the Western Pacific Region (WPRO), providing an empirical timeline of variant propagation.

5.3 Clustering Performance Metrics (MNIST Machine Learning)

The unsupervised K-Means algorithm successfully organized the unlabeled pixel-intensity distributions based purely on statistical coordinate distances. Following the execution of the mode-mapping alignment, the model's classification profile was rigorously quantified using a macro-averaged F1 score:

$$\text{Recorded Macro F1-Score} = 0.8628$$

An F1 score of 0.8628 (86.28%) demonstrates that an unsupervised distance-based clustering routine can effectively map high-dimensional structural topologies into highly accurate semantic classifications, even without receiving training target constraints.

6. Conclusion

6.1 Project Key Takeaways

This project successfully integrated multi-source data engineering workflows with mathematical evaluation models. The foundational phases demonstrated the high efficiency of vectorization within the Pandas ecosystem, proving that multi-million row arrays can be cleaned, type-cast, dynamically sliced, and cross-tabulated with minimal computational latency. The secondary phase highlighted the predictive capacity of basic distance-based clustering algorithms when analyzing flat arrays of raw numerical intensity features.

6.2 Justification of Findings

The empirical conclusions drawn in this report are mathematically justified by the programmatic outputs. The discovery of the peak transmission date (January 30, 2022) is verified by global cross-sectional summations, matching established historical epidemiologic variations. Furthermore, the K-Means clustering model's high macro F1 score of 0.8628 proves that unsupervised spatial partitions match structural labels with strong alignment, validating the use of mathematical models for post-hoc sorting.

6.3 Recommendations for Future Work

While descriptive time-series slicing provides solid historical insights, future expansions should implement predictive analytics. It is highly recommended to deploy autoregressive statistical frameworks (such as ARIMA) or machine learning architectures (such as Facebook Prophet) to forecast trend movements.

Additionally, to expand the digit pipeline from basic numerical vector distance sorting into true computer vision and spatial feature extraction, future iterations should transition from unsupervised clustering to supervised Deep Learning techniques—specifically **Convolutional Neural Networks (CNNs)**. This change would enable the model to actively process spatial hierarchies and edges within visual grids, elevating classification performance from 86.28% to optimal accuracy thresholds (>99%).

7. APPENDICES

To ensure perfect execution replicability across computational environments, the code base requires the following precise version dependencies:

- Language Environment: Python 3.8+
- Core Data Manipulation: pandas (version 1.3.0+)
- Mathematical Libraries: numpy (version 1.21.0+), scipy (version 1.7.0+)
- Machine Learning Framework: scikit-learn (version 0.24.0+)
- External Data Access: gdown (version 4.4.0+)
- Data set used : https://drive.google.com/drive/folders/1gieHICVDBbUKMZiSF4YRQMAJic-w50JM?usp=sharing
- Source code (.ipynb) : https://github.com/MayuukhU15/Time-Series-Analysis/blob/main/06_Time_Series_Analysis_of_COVID_19_Data_Summer_2026.ipynb

1. Abstract

This project presents a comprehensive time-series analysis of the global COVID-19 pandemic using the WHO daily reporting dataset, complemented by an application of unsupervised machine learning on image data. The primary goal was to extract actionable insights from raw temporal data by utilizing Python and the Pandas library for data wrangling, cleaning, and aggregation. Key findings include identifying the top five most affected nations, pinpointing the date of the highest global daily spike (January 30, 2022, with ~8.4 million cases), and visualizing the shifting regional epicenters through quarterly cross-tabulations. In addition to tabular data analysis, the project explored pattern recognition by applying the K-Means clustering algorithm to the MNIST handwritten digit dataset. The model successfully segmented the image features without prior label exposure, achieving a macro-averaged F1 score of 0.8628 after algorithmic label mapping. Ultimately, this internship project demonstrates a complete, end-to-end data science pipeline, bridging the gap between raw data ingestion and robust performance evaluation.

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Topics covered during the initial two-week training period: *(Note: Please review and adjust these based on your actual first two weeks at IDEAS)*

- Python Programming Fundamentals and Data Structures
- Data Wrangling, Subsetting, and Manipulation using Pandas
- Time-Series Data Handling and datetime object conversions
- Grouping, Aggregation, and Constructing Pivot Tables
- Introduction to Machine Learning pipelines and Scikit-Learn
- Unsupervised Learning concepts and K-Means Clustering implementation
- Model Evaluation Metrics (e.g., F1 Score, label mapping techniques)

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3. Project Objective

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- **Optimize data processing efficiency:** Demonstrate proficiency in memory optimization and dataset management through targeted subsetting, boolean masking, and correct data type assignments.
- **Apply unsupervised machine learning:** Utilize K-Means Clustering on the MNIST dataset to independently segment complex image data based purely on intrinsic pixel patterns.
- **Execute rigorous model evaluation:** Evaluate the performance of the unsupervised model against true labels by developing an algorithmic label-mapping loop and calculating the macro-averaged F1 score.

Here is the fully written, professional content for the remaining four sections of your report. It is carefully structured to meet institutional standards and directly incorporates the data and metrics from your Python notebook analysis.

You can copy and paste this text directly into your [Summer Internship Report 2026.docx].

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7. Appendices

Appendix A: Technical Pipeline and Execution Sequence

The following workflow chart illustrates the modular script execution pipeline applied uniformly across the programmatic environment:

[Raw WHO CSV Ingestion via gdown]

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[Feature Subsetting & Copying]

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[pd.to_datetime Numerical Casting]

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[Boolean Mask Range Filtering] ───► [Quarterly / Monthly Pivot Extraction]

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[MNIST Array Ingestion]

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[K-Means Cluster Fitting (K=10)]

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[scipy.stats.mode Label Mapping]

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[Macro F1 Classification Evaluation]

Appendix B: Source Environment Specifications

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